Using Machine Learning to Detect Autism-Related Mutations in Noncoding Regions

Jingyi Han[1], Aziz Zafar[2], Audrey Kris[3], Yufeng Shen[2]

[1]The Peddie School, [2]Columbia University Department of Biomedical Informatics, [3]Colgate University

Introduction

Noncoding DNA sequences are the regions of the genome that do not directly encode proteins. Instead, they include regulatory elements which interact with the coding regions to control their expression. Mutations in these regions can still manifest in disorders such as autism.

Because there is little data on these regions, it is hard to directly predict mutations. Large language models (LLMs) like our project GenoSCOPE (Genomic Selection Coefficient Optimization via Probabilistic Estimation) trained on DNA can bypass this restriction in order to predict mutations in noncoding regions.

Background

Bayesian analysis is a method of incorporating prior knowledge into probabilities. LLMs are used to calculate an informed prior probability distribution (the “prior”) used to ultimately predict the constraint of a gene, or how much it tolerates mutation. Constraint could potentially be used to predict the effect of certain mutations, including autism and other disorders.

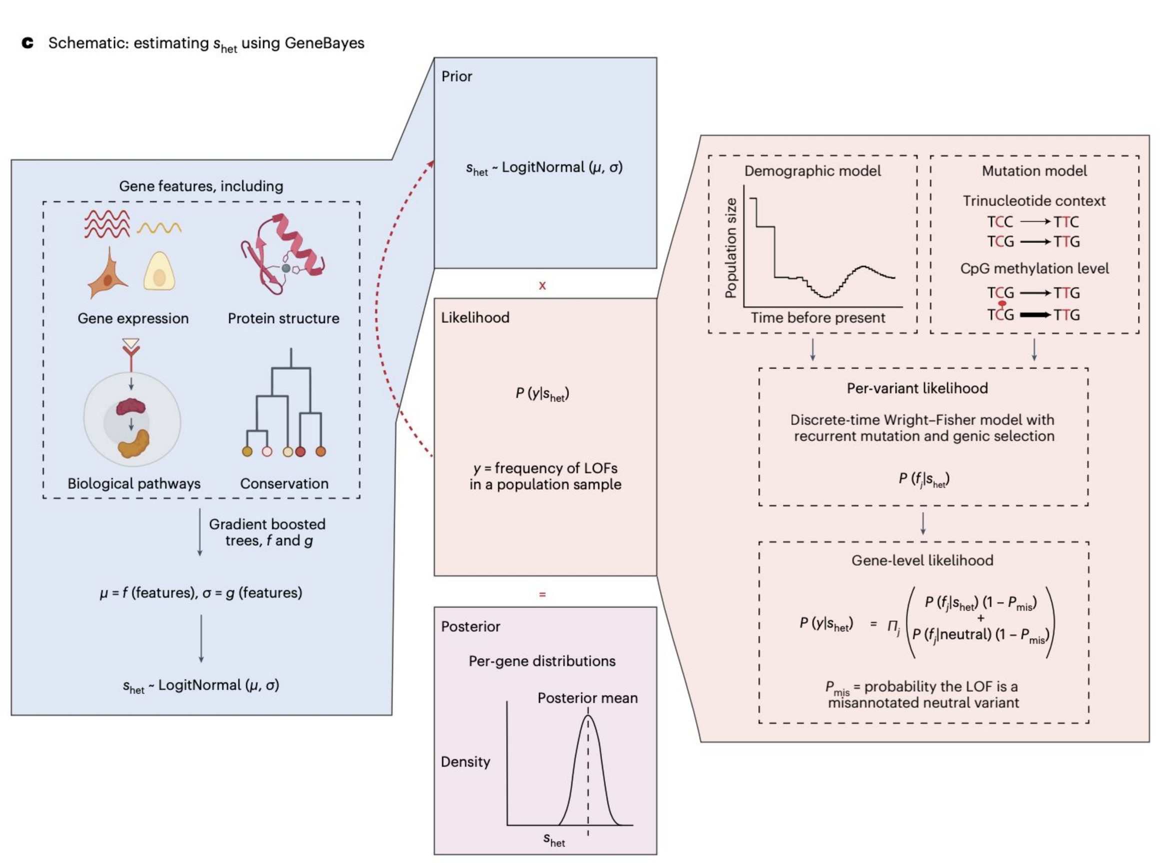

Figure 1: A Bayesian approach to gene expression predictions

For this purpose, the prior used is the number of observed mutations divided by the number of expected mutations, or o/e, which is related to constraint: if there are far fewer observed mutations than expected, those mutations would be far more disruptive to the gene, giving it a higher constraint.

Research Question

How can Large Language Models be used to predict mutations in noncoding regions of DNA?

Methodology

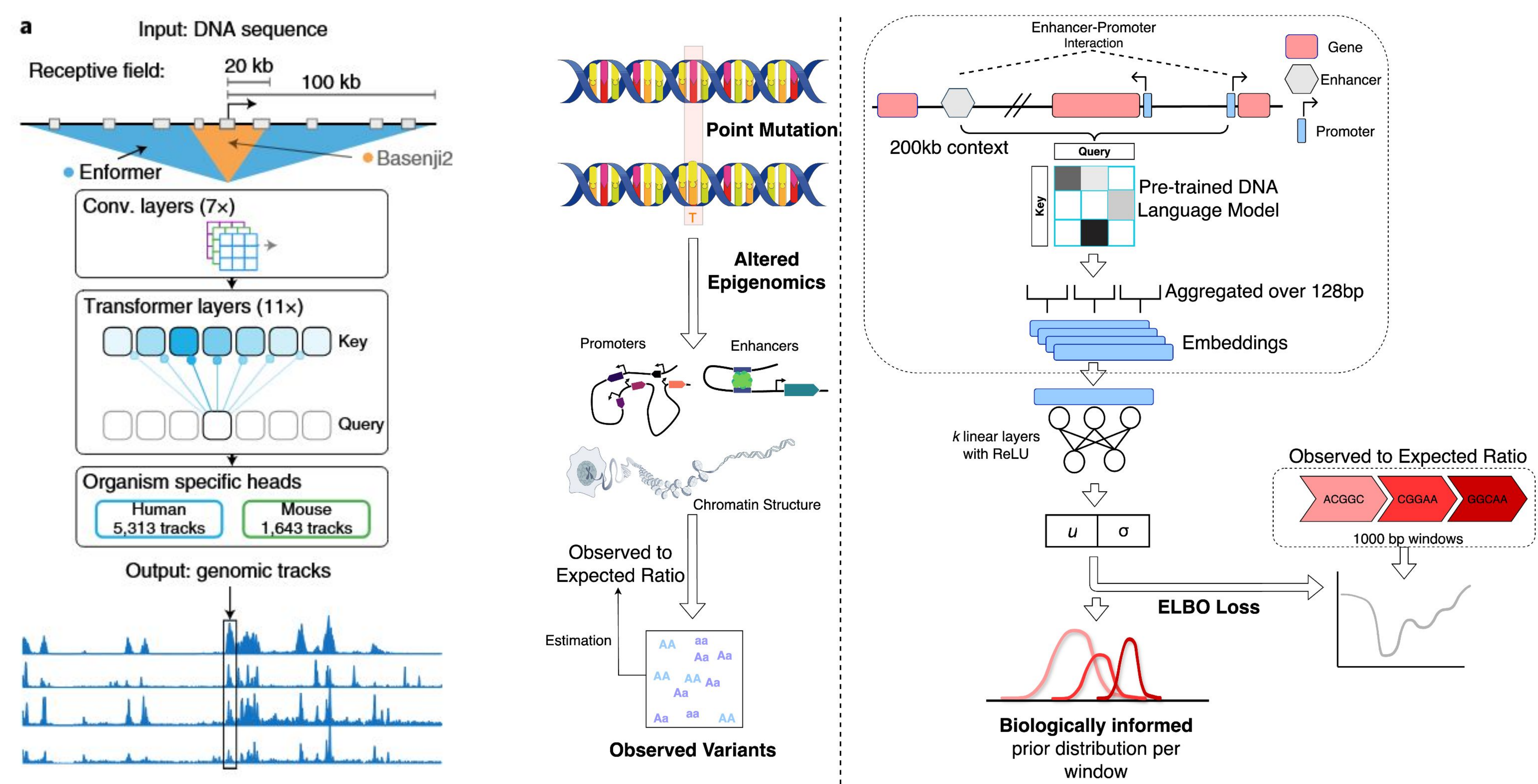

Figure 2: Enformer architecture and outputted tracks, **Figure 3:** GenoSCOPE Workflow

GenoSCOPE is trained on a dataset generated by Enformer, which analyzes DNA and outputs epigenomic “tracks”: data describing its features such as gene expression or chromatin openness. The data, grouped in 1000 base-pair windows, is labelled with o/e values calculated with Gnocchi.

This model predicts o/e using a regularized linear regression that reduces both the magnitude and number of coefficients, creating a simpler algorithm that is less tailored to specific training data. While previous models struggled with overfitting and bad performance on unseen data, regularization prevents this.

Model Agnostic Prediction Interval Estimator (MAPIE) is then used to achieve a probability distribution from the point predictions, which provides the prior for the Bayesian estimation.

References/Acknowledgements

[1]: Zeng, T., Spence, J.P., Mostafavi, H. et al. Bayesian estimation of gene constraint from an evolutionary model with gene features. Nat Genet 56, 1632–1643 (2024). https://doi.org/10.1038/s41588-024-01820-9
[2]: Chen, S., Francioli, L. C., Goodrich, J. K. et al. A genomic mutational constraint map using variation in 76,156 human genomes. Nature, 625(7993), 92–100. (2024). https://doi.org/10.1038/s41586-023-06045-0

Thank you to the Peddie EXP program, Ms. Wolfe, Dr. Shen, Aziz, and Audrey for the support!

Results

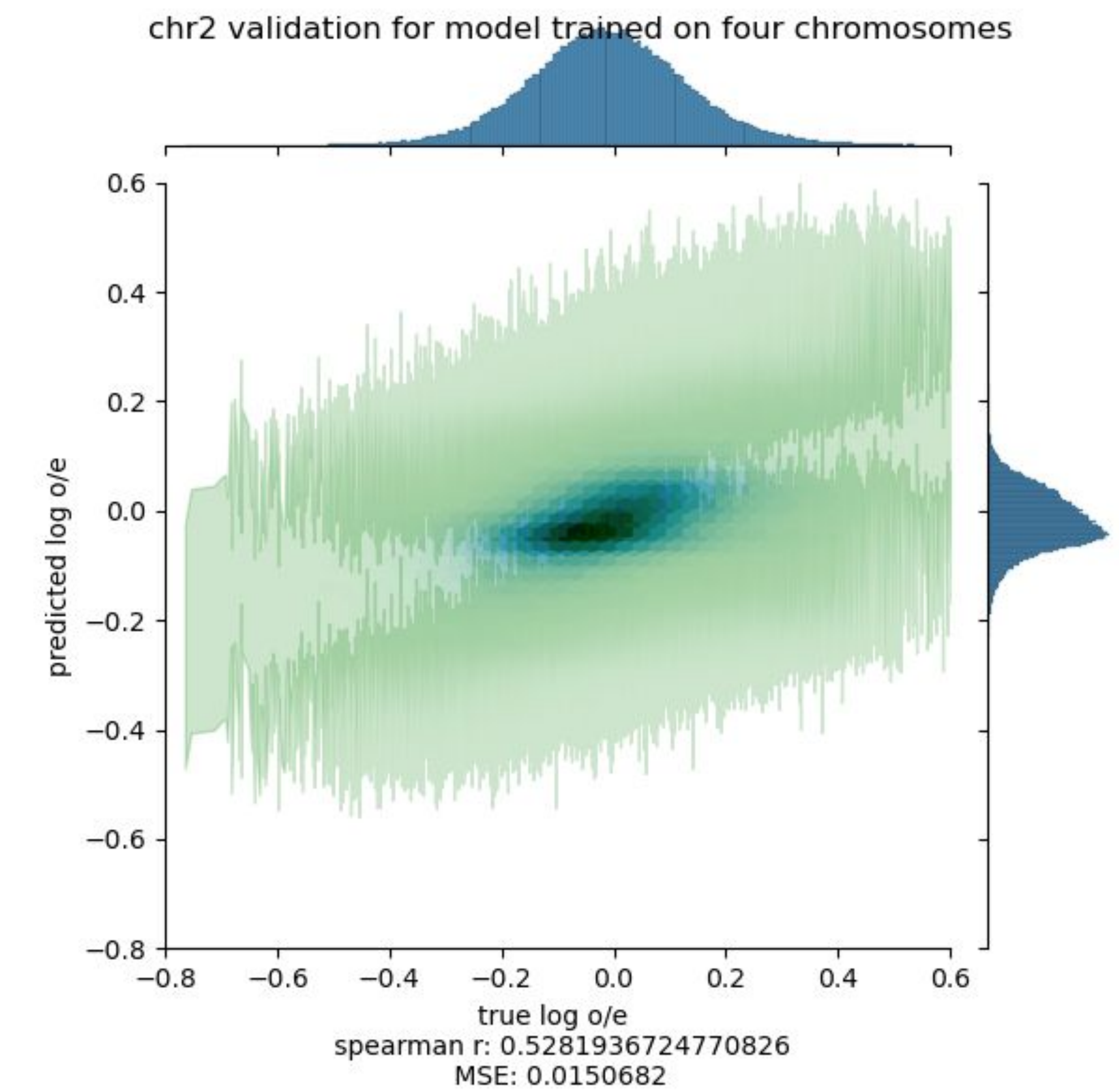

Figure 4: GenoSCOPE’s predicted o/e plotted against true values

The blue heatmap shows the concentration of the predicted points, while the transparent green shows the uncertainty distribution for each predicted point.

GenoSCOPE is effective at predicting o/e for unseen data grouped in 1000 base-pair windows without overfitting.

Conclusion

DNA LLMs show great promise in zero-shot predictions such as o/e for noncoding regions, working with mutations without extensive data and documentation.

Futures:

- Reduce window sizes from 1000 bp to 500 and eventually 100 bp
- Calculate o/e effectively for smaller window sizes
- Use o/e predictions as a prior to predict gene constraint

Jingyi Han   jhan-26@peddie.org